IMO Training – Number Theory
Level 1 Assignment 2

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Instructions:

- Write your solutions legibly on separate sheets of paper.
- Each question carries 7 marks.
- Complete solutions with justified arguments are required for full credit in every question.
- Both paper and electronic copies are accepted.
- Deadline for submission: 16 November, 2019

1. Find all $x, y \in \mathbb{N}$ s.t. $x + y + 1 \mid 2xy$ and $x + y - 1 \mid x^2 + y^2 - 1$.
2. Are there infinitely many pairs of positive integers (m, n) s.t. $m \mid n^2 + 1$ and $n \mid m^2 + 1$?
3. Let $a_0 = 4$ and define sequence $a_n = a_{n-1}^2 - a_{n-1}$ for each $n \in \mathbb{N}$.
   (a) Prove that there are infinitely many primes which are factors of at least one term in the sequence.
   (b) Are there infinitely many primes which are factors of no term in the sequence?
4. Suppose that p is a prime and that there are different $u, v \in \mathbb{N}$ s.t. p^2 is the mean of u^2 and v^2. Prove that $2p - u - v$ is either a square or twice a square.
5. Determine whether there exist $n \in \mathbb{N}$ s.t. $\displaystyle\sum_{n+1}^{2n} n^3$ is a square.
6. Let p be an odd prime. How many non-empty subsets of $1, 2, 3, \cdots, p-2, p-1$ have a sum which is divisible by p?

<u>Classwork</u>

Primitive Roots Notes Examples 2, 9-11, 21-23, 44-47, 51.